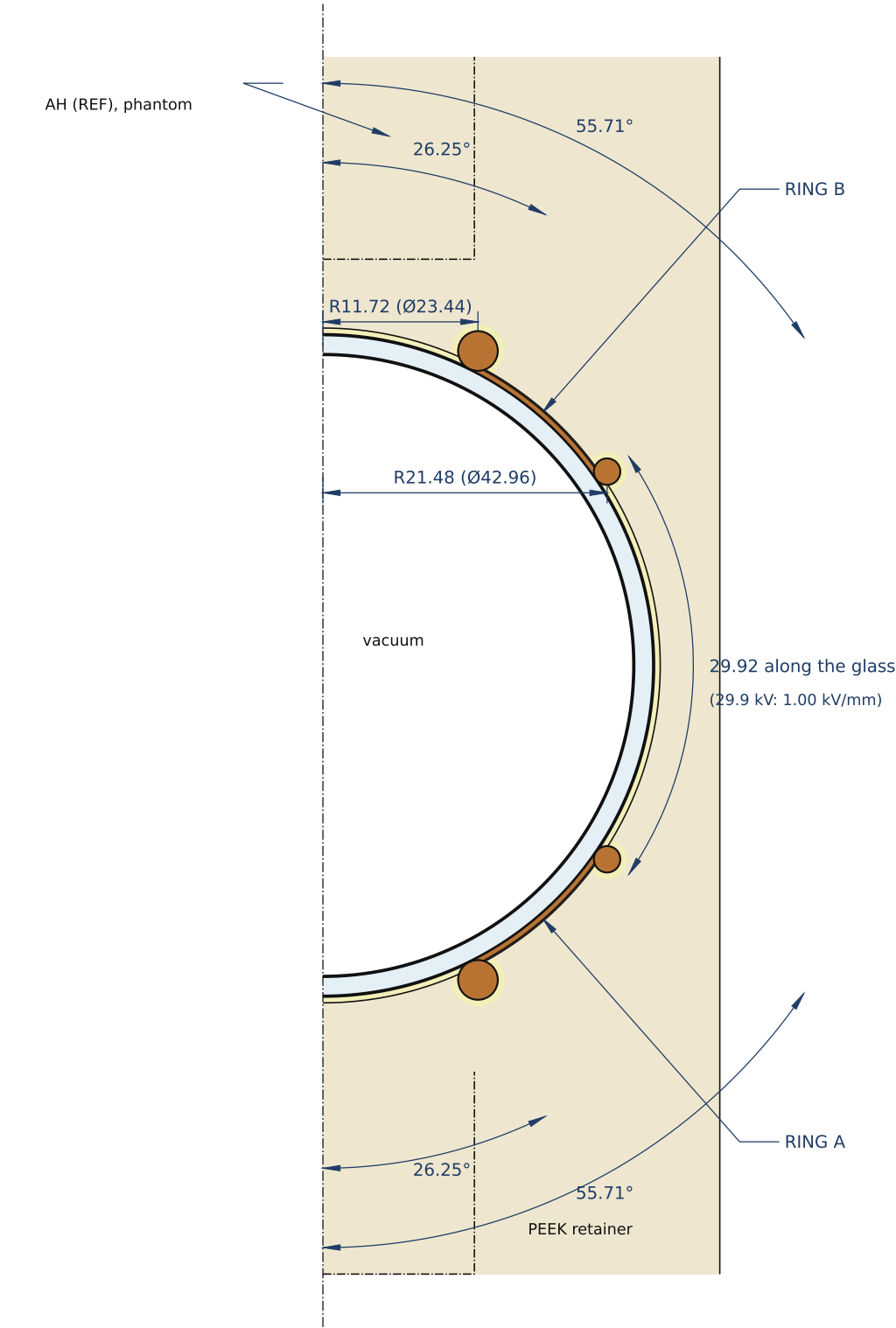
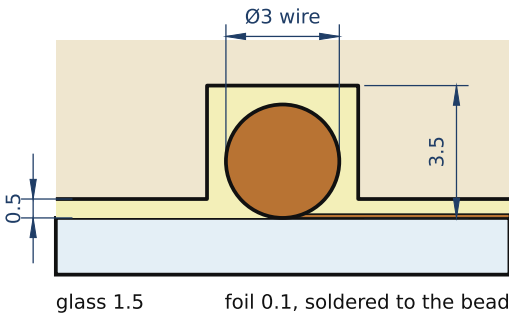


A HALF-SECTION ON THE AXIS (2:1)

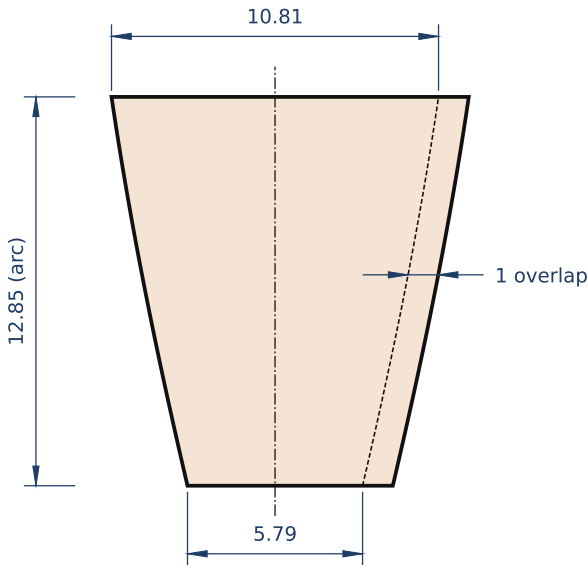
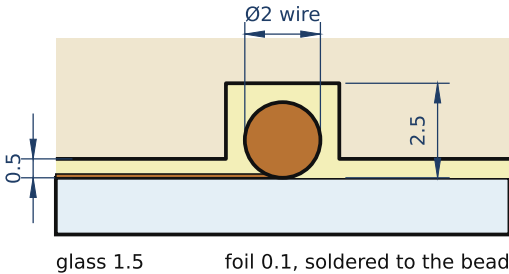
rings A and B mirror the equator; angles from the axis



B POLAR BEAD, Ø3 (5:1)



C EQUATORIAL BEAD, Ø2 (5:1)



D GORE, FLAT (4:1): 12 per band, 24 in all

width = $2\pi \cdot 25 \cdot \sin \theta / 12$ along the arc; the edge θ at the narrow end

RINGS A AND B: band 12.85 along the glass, 13.1 cm² each; the gap 29.92

feature (each ring)	polar edge	equatorial edge
edge angle from the axis	26.25°	55.71°
bead's contact on the glass	r 11.06 / z ±22.42	r 20.65 / z ±14.08
bead's centre	r 11.72 / z ±23.77	r 21.48 / z ±14.65
bead: Cu wire, ring centre-line	Ø3 wire, Ø23.44	Ø2 wire, Ø42.96
wire per ring (cut, close, solder)	73.6	135.0
groove in the PEEK (depth)	3.5	2.5
peak field in the gel (record)	4.85 kV/mm	4.36 kV/mm

NOTES

1. Material: foil Cu-ETP (CW004A), annealed, 0.10 thick; bead rings Cu-ETP wire. Glass: the vessel as supplied (borosilicate, OD 50, wall 1.5), clean and degreased.
2. Cut 12 gores per band (D); burnish onto the glass from the polar edge, each overlapping its neighbour by 1; solder the overlaps flat. No adhesive film under the foil: the gel bonds and fills it.
3. Close each bead ring by a butt solder joint, filed round; solder it along the foil's edge, centred on the edge angle (A), touching the glass. Smooth every joint: no point or burr above the bead's radius.
4. The leads: PTFE-insulated HV wire soldered to the equatorial bead, out through the PEEK and the G10 coupler.
5. The PEEK retainer: unfilled, annealed stock; the pocket 0.5 over the glass and the grooves (B, C) machined to size; degassed silicone gel vacuum-cast into the pocket and the grooves, void-free.
6. Ring A is ring B mirrored in the equator. Test before potting: continuity; after: hold-off per docs/bench-test-rings.md phase 1 (c).

DCCREG turbine · the hub: the field's rings on the vessel

RINGS A AND B, COPPER, BEADED

Drawing no.

Cu-ETP foil 0.10 + wire; in gel under PEEK

DCCREG-HUB-201

Scale: 2:1 (A), 5:1 (B, C), 4:1 (D) Units: mm

Rev A (draft) Sheet 1/1 A3

Drawn: generated from sim/hub_rings_build.py

Date: 2026-10-09